

Let's Encrypt HTTPS — Status Summary

Revision: 1 **Last modified:** 2026-07-01T12:15:00Z **Status:**
Companion summary of [Status.md](#) (§11.4.56 two-audience).

Page 1 — For the operator / stakeholders (plain language)

We are adding **automatic HTTPS** to the proxy using **Let's Encrypt** — free, auto-renewing TLS certificates — with **Caddy** doing the certificate work in-process (no scripts or cron to babysit) and the **DNS-01** method so it works even without opening ports on the host.

What works today:

- The three setup decisions are locked in (Caddy, DNS-01, test-first then defer the real domain).
- A **certificate checker** is built and proven: it can tell whether a resulting certificate is still valid, how many days are left, whether it covers the right hostname, and whether it was issued by the expected authority — all offline, and it is self-tested so it cannot lie (a deliberately-broken certificate is always caught).

What's still pending / needs you:

- Issuing a real certificate end-to-end (first against a local test server, then Let's Encrypt **staging**) — the next automated step is the local hermetic test.
- **Going live on the real domain** needs you to provide a DNS provider API token and give the go-ahead. Until then nothing touches production.

Bottom line: the foundation and the safety-checker are done and proven; certificate issuance and the real-domain go-live are the remaining steps, the last one waiting on your go-ahead.

Page 2 — For software engineers

- **Task:** #59 Let's Encrypt HTTPS + auto-renewal + rotation (feature workstream, §11.4.167).
- **Phase 0 (PASS):** operator decisions captured via §11.4.66 — Caddy auto-HTTPS / DNS-01 / hermetic+staging-first, defer real domain. Design LETSENCRYPT_HTTPS.md + plan_PLAN.md @ fddd25c.
- **Phase 1 cert-analyzer (PASS):** tests/letsencrypt/cert_analyzer.sh — 5 pure, now-seam-deterministic (§11.4.50 CERT_ANALYZER_NOW_EPOCH) functions; SAN match is exact-token + single-leading-wildcard (no substring, mirrors_code_in); chain check is issuance-only (openssl verify -no_check_time), orthogonal to expiry. Self-validated §11.4.107(10): cert_analyzer_selftest.sh 37/37; tests/regression/cert_analyzer_selfvalidation_test.sh GREEN+RED_MODE=1 PASS; §1.1 mutation (SAN exact→substring) → guard FAIL, restore byte-identical md5 a06e0f89ee9993fe5de368be852e4eff. Fixtures: public *.pem tracked, *.key/*.csr/*.srl gitignored (§11.4.10), regen via fixtures/gen_fixtures.sh (§11.4.77).
- **Pending (autonomous):** custom Caddy DNS-01 image (§11.4.76 containers submodule, rootless §11.4.161), Phase 2 compose+secret+volume wiring, Phase 3 Pebble hermetic issuance (anti-bluff core), Phase 5 renewal/rotation sim.
- **OPERATOR-BLOCKED:** Phase 4 LE-staging (real DNS-01 token §11.4.10), Phase 6 prod cutover (real-domain go-live gate, design §9).
- **Guard wiring:** cert_analyzer_selfvalidation_test.sh to be registered in run-tests.sh test_regression_guards() (§11.4.135) with the LE lane commit.
- **Anti-bluff:** every non-PASS row is honestly PENDING/ OPERATOR-BLOCKED (§11.4.6) — no metadata-only PASS; the analyzer is ready to assert the output of the issuance phases when they land.